॥वसुधैव कुटुम्बकम्॥

SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Symbiosis International (Deemed University)

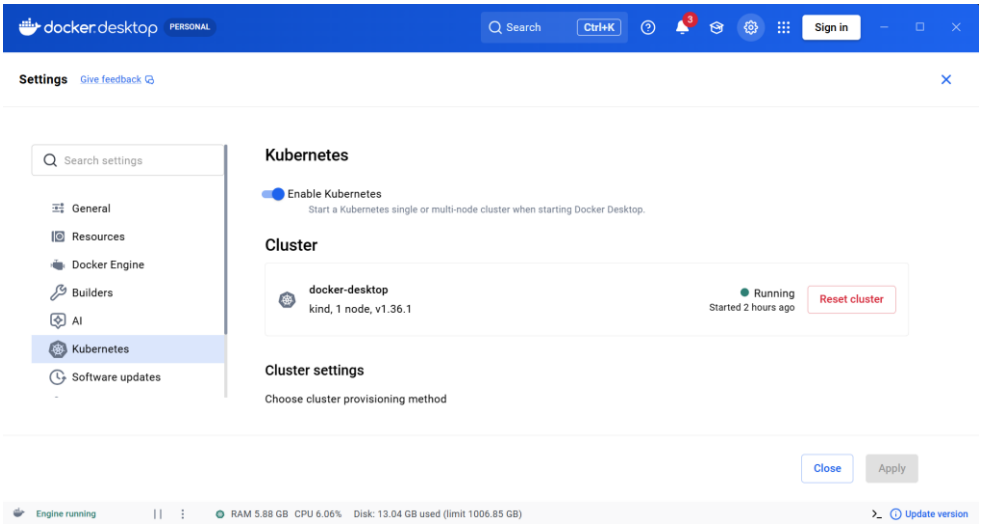
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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Assignment No. 12

Subject:	DevOps Lab
TCODE:	TE7950
Name of Student	**Yash Patel**
PRN No.	**23070122235**
Branch	CSE, Batch (2023-27), L4
Academic Year & Semester	25 – 26, Sem VII
Date of Performance	23rd July, 2026
Title of Assignment:	Kubernetes Pod & Deployment
Theory:	Kubernetes is an open-source container orchestration platform used to automate the deployment, scaling, and management of containerized applications. A Pod is the smallest deployable unit in Kubernetes and contains one or more containers that share the same network and storage resources. A Deployment manages Pods by maintaining the desired number of replicas, automatically replacing failed Pods, and supporting updates without downtime. Kubernetes simplifies application management by ensuring high availability, scalability, and efficient resource utilization across a cluster. **Commands Used** • Checked the Kubernetes client version: kubectl version --client • Verified the Kubernetes cluster information: kubectl cluster-info • Displayed the available cluster nodes: kubectl get nodes • Created the Kubernetes Deployment from the YAML manifest: kubectl apply -f web-deployment.yaml • Listed the running Pods: kubectl get pods • Deleted the Kubernetes Deployment and its Pods: kubectl delete -f web-deployment.yaml

Implementation:	**Step 1: Enable Kubernetes in Docker Desktop** Enabled Kubernetes — Enabled Kubernetes in Docker Desktop and verified that the local Kubernetes cluster was successfully initialized. **Step 2: Verify kubectl** Verified Kubernetes Cluster — Verified that the local Kubernetes cluster was running successfully using kubectl cluster-info and confirmed that the control-plane node was in the Ready state using kubectl get nodes. **Step 3: Create the Kubernetes Deployment YAML** Created Kubernetes Deployment Manifest — Created the web-deployment.yaml file inside the k8s-manifests directory, defining a Kubernetes Deployment with two replicas of the Flask web application using the Docker image my-flask-app:v2. **Step 4: Apply the Deployment Manifest** Applied Deployment Manifest — Applied the Kubernetes Deployment manifest using kubectl apply, creating the Deployment resource in the local Kubernetes cluster. **Step 5: Verify the Pods** Verified Running Pods — Verified that the Kubernetes Deployment successfully created two running Pods, each hosting an instance of the Flask web application. **Step 6: Clean Up** Deleted Kubernetes Deployment — Removed the Kubernetes Deployment and the Pods it created using the kubectl delete command, cleaning up the local Kubernetes cluster. 

```
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\yashp>kubectl version --client
Client Version: v1.36.1
Kustomize Version: v5.8.1

C:\Users\yashp>kubectl cluster-info
Kubernetes control plane is running at https://127.0.0.1:63633
CoreDNS is running at https://127.0.0.1:63633/api/v1/namespaces/kube-system/services/kube-dns:dns/proxy

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.

C:\Users\yashp>kubectl get nodes
NAME                   STATUS   ROLES           AGE     VERSION
desktop-control-plane   Ready    control-plane   2d13h   v1.36.1

C:\Users\yashp>
```

```
C:\Users\yashp>cd "C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects"

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects>mkdir "Project 12"
A subdirectory or file Project 12 already exists.

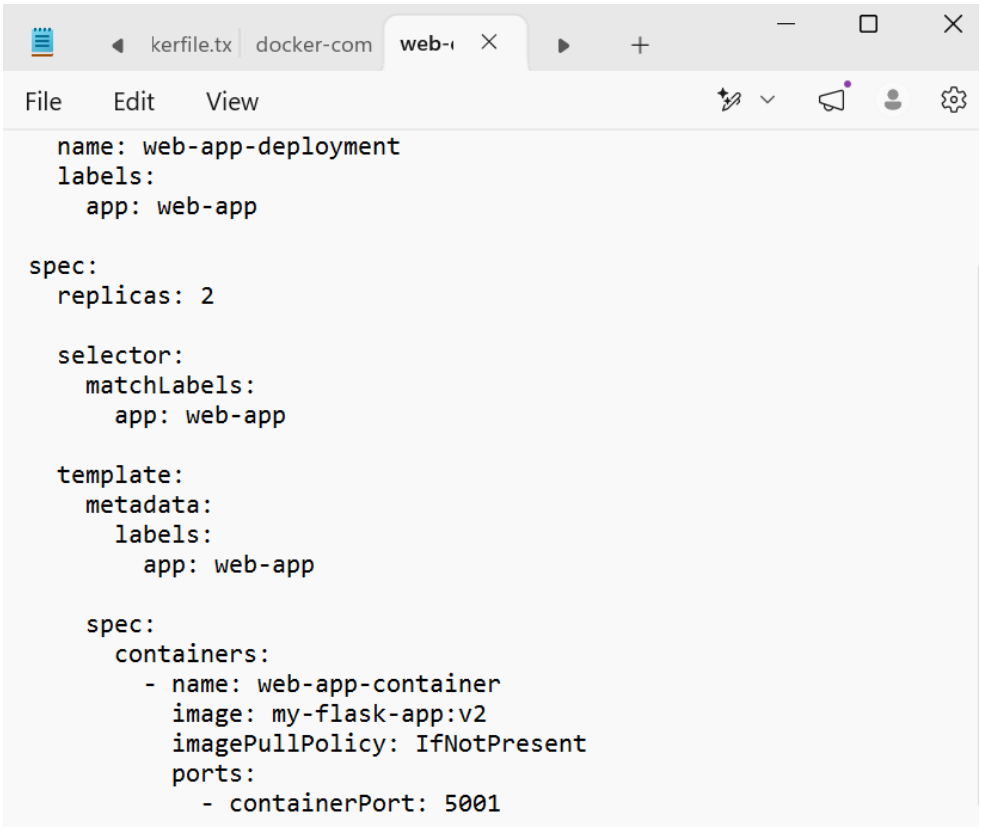
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects>cd "Project 12"

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 12>mkdir k8s-manifests

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 12>cd k8s-manifests

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 12\k8s-manifests>cd
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 12\k8s-manifests

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 12\k8s-manifests>
```




```yaml
  name: web-app-deployment
  labels:
    app: web-app

spec:
  replicas: 2

  selector:
    matchLabels:
      app: web-app

  template:
    metadata:
      labels:
        app: web-app

    spec:
      containers:
        - name: web-app-container
          image: my-flask-app:v2
          imagePullPolicy: IfNotPresent
          ports:
            - containerPort: 5001
```

```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 12\k8s-manifests>type web-deployment.yaml
apiVersion: apps/v1
kind: Deployment

metadata:
  name: web-app-deployment
  labels:
    app: web-app

spec:
  replicas: 2

  selector:
    matchLabels:
      app: web-app

  template:
    metadata:
      labels:
        app: web-app

    spec:
      containers:
        - name: web-app-container
          image: my-flask-app:v2
          imagePullPolicy: IfNotPresent
          ports:
            - containerPort: 5001
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 12\k8s-manifests>
```

```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 12\k8s-manifests>kubectl apply -f web-deployment.yaml
deployment.apps/web-app-deployment created

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 12\k8s-manifests>
```

```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 12\k8s-manifests>kubectl get pods
NAME                                      READY   STATUS    RESTARTS       AGE
cloud-negotiation-agent-dc75b5cbc-c7h6j   1/1     Running   1 (145m ago)   2d7h
cloud-negotiation-agent-dc75b5cbc-rghjf   1/1     Running   1 (145m ago)   2d7h
cloud-negotiation-agent-dc75b5cbc-wwpzg   1/1     Running   1 (145m ago)   2d7h
web-app-deployment-848b75c97-6kvcx        1/1     Running   0              71s
web-app-deployment-848b75c97-dhdv8        1/1     Running   0              71s

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 12\k8s-manifests>
```

```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 12\k8s-manifests>kubectl get pods
NAME                                      READY   STATUS        RESTARTS       AGE
cloud-negotiation-agent-dc75b5cbc-c7h6j   1/1     Running       1 (146m ago)   2d7h
cloud-negotiation-agent-dc75b5cbc-rghjf   1/1     Running       1 (146m ago)   2d7h
cloud-negotiation-agent-dc75b5cbc-wwpzg   1/1     Running       1 (146m ago)   2d7h
web-app-deployment-848b75c97-6kvcx        1/1     Terminating   0              2m12s
web-app-deployment-848b75c97-dhdv8        1/1     Terminating   0              2m12s

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 12\k8s-manifests>kubectl get pods
NAME                                      READY   STATUS    RESTARTS       AGE
cloud-negotiation-agent-dc75b5cbc-c7h6j   1/1     Running   1 (147m ago)   2d7h
cloud-negotiation-agent-dc75b5cbc-rghjf   1/1     Running   1 (147m ago)   2d7h
cloud-negotiation-agent-dc75b5cbc-wwpzg   1/1     Running   1 (147m ago)   2d7h

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 12\k8s-manifests>
```